



TrueVision

Team 3

Members: Yoonseong Kim, Minsu Pak

Problem

- AI generate images can be seen in great quantity on the internet
- Sometimes bad actors will use this technology to fulfill their agenda.
 - This can be anything from deep fakes or AI generated images to tell fake news or selling AI work as human work.
- In the beginning, it was possible to tell what images have been generated by AI but now it has become increasingly difficult to do so.

Solution

- Develop a web application that uses AI to determine if the images have been generated by AI
- Our platform will also provide users with the ability to see how the AI is determining an image is AI using ViT's attention.
- We also offer an API for end users to use our models, which similar platforms we found did not offer



Main use cases

Main Use Cases

1: Submit Image

Actor: User

Priority: Essential

- User selects an image
- Image sent to the backend
- Rejected if not an image file

2: Evaluate Image

Actor: Backend

Priority: Essential

- Backend receives file
- Checks the file is an image
- Pass image through the AI model
- Send results to frontend/user

3: Select Model

Actor: User

Priority: Essential

- User can pick a model
- Choice is saved on the frontend
- Choice is limited if the user is not paid

4: Purchase Paid Subscription

Actors: User, backend, database

Priority: Expected

- User is logged in
- User clicks to purchase paid plan
- Backend receives needed data & authenticates user
- Database is updated

5: Manage API Key

Actors: User, backend, database

Priority: Medium

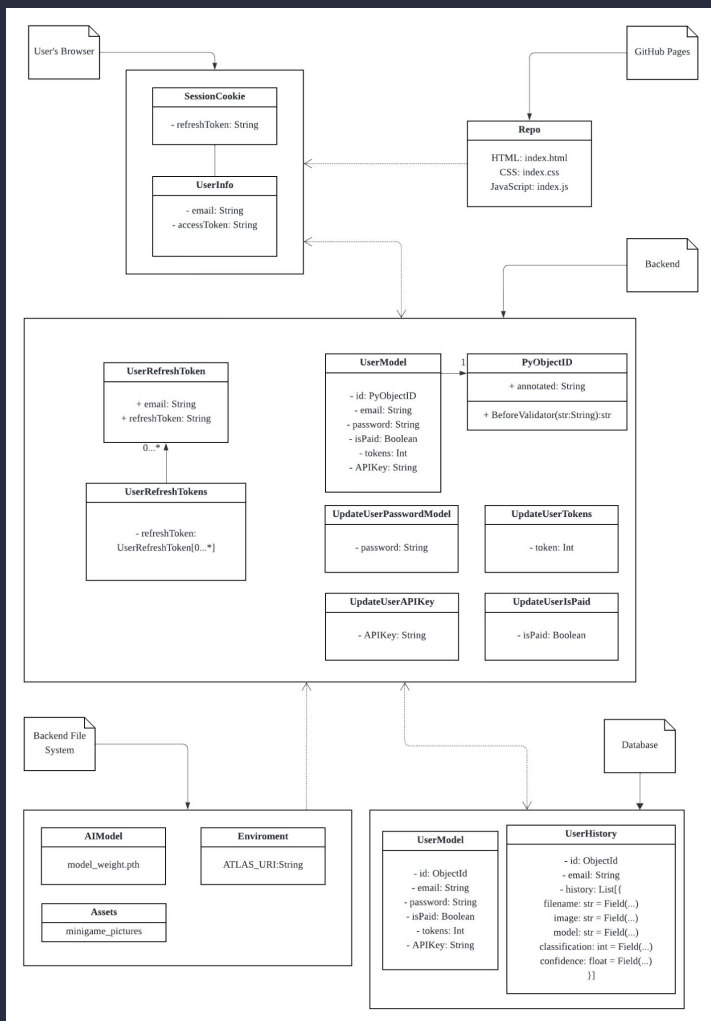
- User is logged in and is a paid subscriber
- User can manage API key on the Account Settings page
- User can regenerate a new API Key
- Backend generates new key
- Updates database

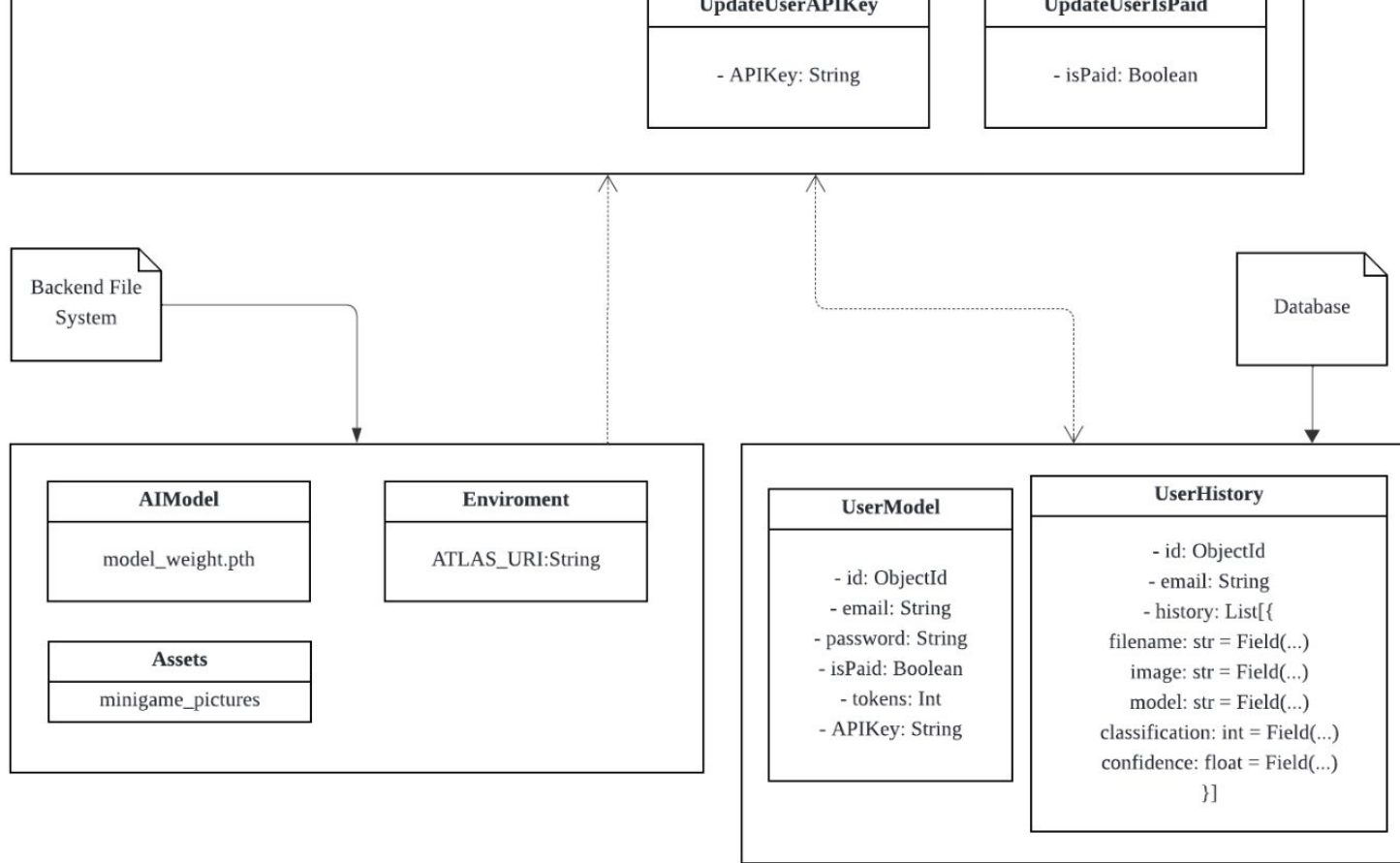
All complete!! :)



Project Update: UML Data Diagram









Added Features

New Features

Save user analysis history

Actors: User, backend, database

- User has an account and analyses an image
- Image sent to the backend
- Backend analyses image
- Analysis results saved as history on the database & image on the backend
- User's history displayed on the results page
- History persists through browser & browser session

New Features

Image guessing minigame

Actors: User, backend

- The user is presented the option to pick one of three models
- The backend presents an image to the user
- The user guesses if the image is AI-generated or human original
- The backend then instructs the model to make a guess
- The outcomes are sent back to the front end for local statistics tracking (Only persistent for that session)



Project Update: New Requirements

Section 12

Req 12-1: The page shall offer 3 models to play against

Req 12-2: The page shall display the challenge image

Req 12-3: The page shall provide buttons to select if an image is AI or Human

Req 12-4: The page shall notify when the AI is guessing

Req 12-5: The page shall keep track of the number of rounds

Req 12-6: The page shall keep track of the number of user wins

Req 12-7: The page shall keep track of the number of AI wins

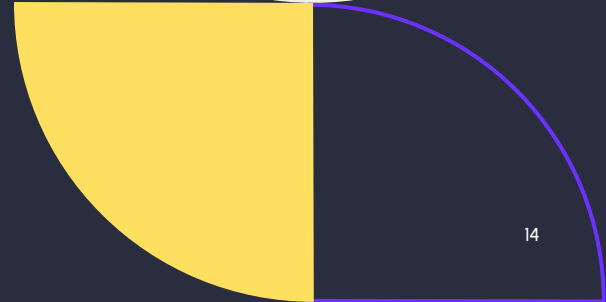
Req 12-8: The page shall display the AI's attention

Req 12-9: The page shall provide a select another model button

Req 12-10: The page shall provide a play again button



Project phase: Launch and go-live



Deployment Methods



Frontend

It is free!



Backend

It is free!



Database

It is free!

TrueVision's AI

A deeper look into our
models

The Datasets

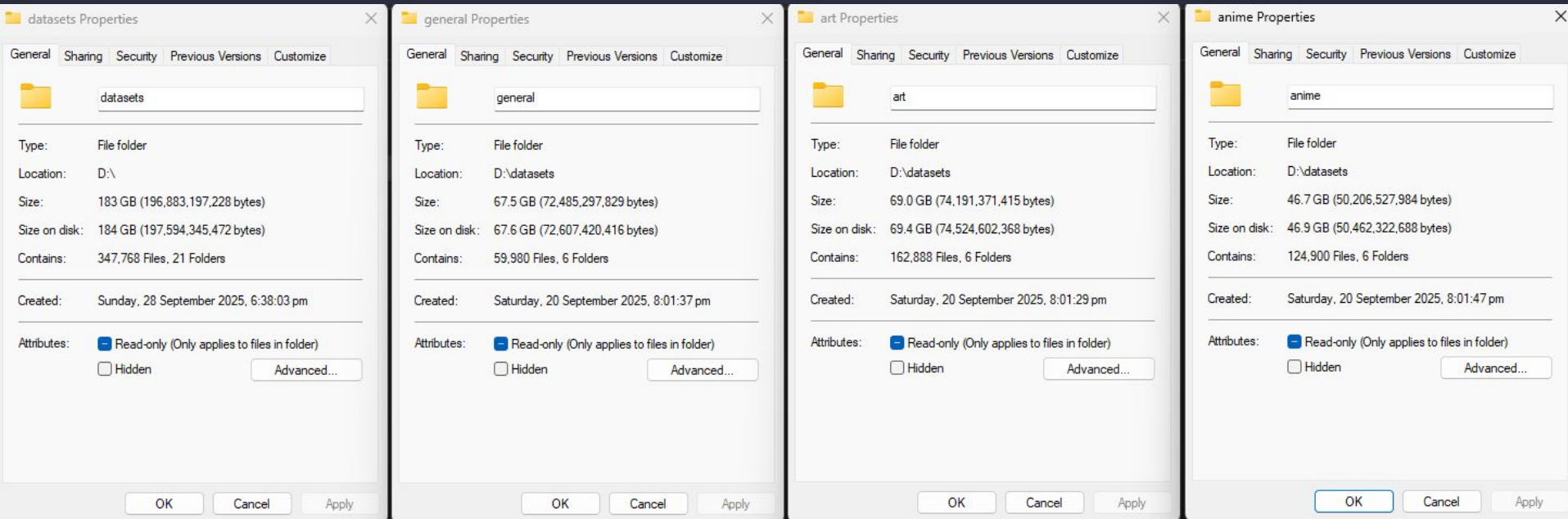


Image Distribution



General



Training: 48,000

Testing: 12,000

Art



Training: 151,000

Testing: 12,000

Anime

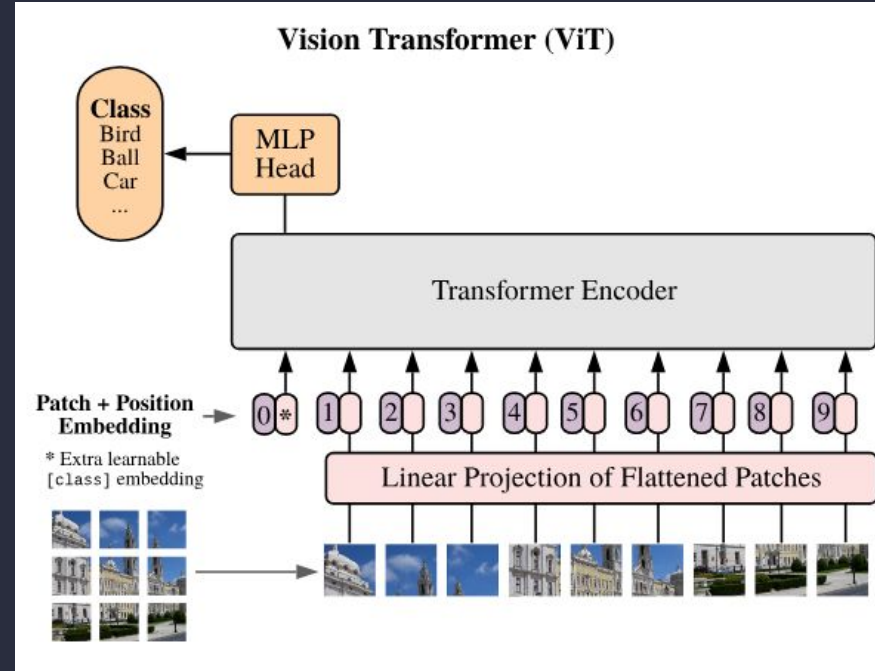


Training: 116,000

Testing: 12,000

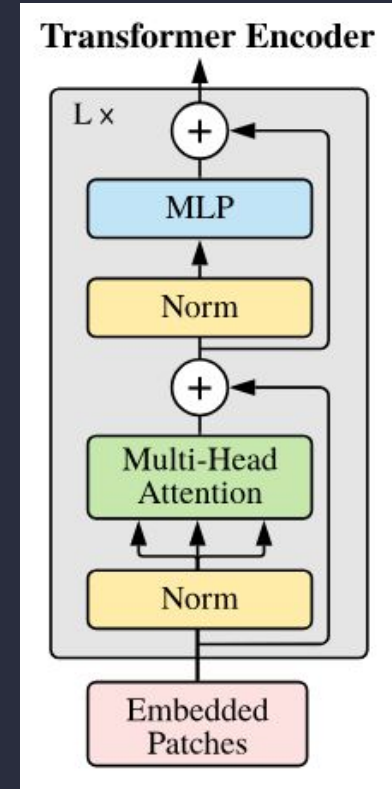
Model Architecture

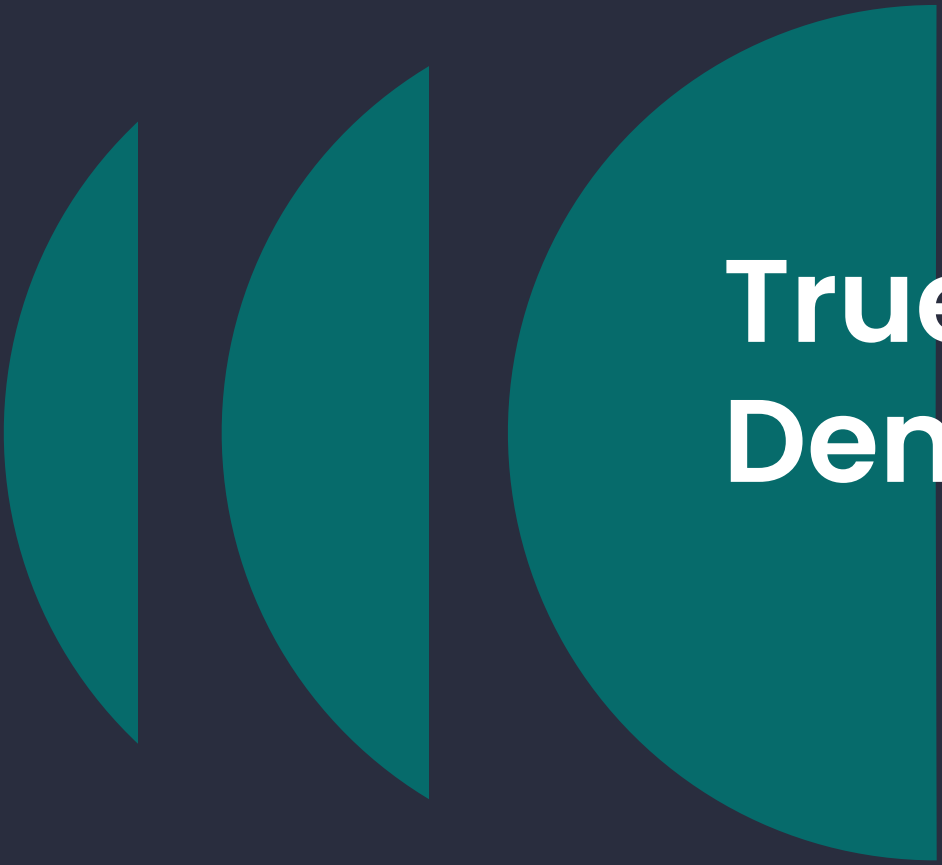
- Uses ViT-B16-224
- Patch size: 16x16
- Patches: $14 \times 14 = 196$ patches
- 12 transformer encoder layers
- Image resized to 672 x 672
- Image divided into 9 parts (224×224 each)
- Takes the average prediction of 9 parts



Model Architecture

- Uses ViT-B16-224
- Patch size: 16x16
- Patches: $14 \times 14 = 196$ patches
- 12 transformer encoder layers
- Image resized to 672×672
- Image divided into 9 parts (224×224 each)
- Takes the average prediction of 9 parts





TrueVision Demo Time

This one first

<https://220.126.157.140:10000/docs>

This one second / Our Frontend

<https://yoonseongkim2000.github.io/CSE-416-Team3/>

Contributions

Minsu Pak

- Database & backend-to-database integration
- Account and authentication functionality (front & back end)
- Frontend + GUI styling
- Logo & graphics

Contributions

Yoonseong Kim

- 3 AI Models provided for the TrueVision Project
- Developed Front and Back End
- Retrofitted CSE 316 Project for API Use Demonstration
- Made the Minigame Page
- Stayed out of Minsu's Way in terms of design (!Important)

Q/A